

VANSIKA JHAWAR

+91 9674795110 | E-mail | Portfolio | LinkedIn | GitHub

Machine learning researcher focused on **explainable and trustworthy AI**, applying **interpretability methods (SHAP, LIME)** to security- and safety-critical systems. GATE-qualified in CS and Data Science & AI; incoming NTU MSc student in Computer Control & Automation.

EDUCATION

Nanyang Technological University

MSc in Computer Control & Automation, School of Electrical & Electronic Engineering

Singapore

Incoming, 2026

Kalinga Institute of Industrial Technology

B.Tech (Honours) in Computer Science and Engineering, Minor in Applied Machine Learning

Odisha, India

Aug 2022 – May 2026

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL

AI/ML & Safety: Machine Learning, Deep Learning, TensorFlow, Explainable AI / Interpretability (SHAP, LIME), Computer Vision, Model Evaluation

Tools & Infrastructure: Git, GitHub, Docker, Pytest, REST APIs, AWS, WebRTC

Databases: PostgreSQL, SQLite

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Software Engineering

RESEARCH & PUBLICATIONS

Transparent Intrusion Detection via Explainable LSTM–HMM Hybrid | IEEE-ICCECE 2026 (Accepted)

2026

- Designed a hybrid **LSTM + Hidden Markov Model** intrusion detection system and applied **SHAP and LIME** to render its decisions interpretable for human security analysts – directly addressing the model-transparency and human-oversight problems central to AI safety evaluation.

Vehicular Fog Computing at the IoV–Renewable Energy Nexus | IGI Global, Published 2025

2025

- Lead-authored a chapter on distributed fog-computing architectures for connected, safety-critical vehicular systems.

Balancing Innovation and Ethics in AI-Integrated Social Media Platforms: A Review | IGI Global (Submitted)

2026

- Co-authored a review examining the ethical trade-offs of AI integration in social media platforms, submitted for the volume *Lifespan Approaches to Evidence-Based Digital Psychological Interventions*.

SELECTED PROJECTS

BoneNet – Explainable AI Diagnostic Tool | Python, TensorFlow, SHAP, LIME

2025

- Built a deep learning model for bone-fracture detection from X-rays and integrated **SHAP/LIME interpretability layers** to produce clinician-legible heatmaps – a human-in-the-loop pattern for deploying opaque models in high-stakes decisions.

OceanCleanUp – AI-Driven Marine Debris Detection | Python, Computer Vision, Drones

Ongoing

- Designing a computer-vision and drone-based sensing pipeline to identify and map ocean plastic accumulation zones, enabling targeted, evidence-driven cleanup at scale.

MindScan – Gamified Mental Health Screening | Python, Pygame, SQLite, ML

Mar 2025

- Screened **100+ users** across 25 mental-health/learning-disorder profiles via an embedded ML-driven Ludo game; achieved **90% test coverage** (Pytest) on a **1000+ record** SQLite backend.

Annapurna – IoT Agriculture Platform | Python, IoT Sensors, REST APIs, AWS

Nov 2024

- Real-time monitoring platform bridging **10+ IoT sensors** with precision farming analytics; **90% system reliability** via SDLC-aligned testing.

ACHIEVEMENTS

- **Qualified GATE 2025 & 2026** in both **Computer Science (CS)** and **Data Science & Artificial Intelligence (DA)**.
- **First Runner-Up, IET India Scholarship Award 2025 (National)** – Rs. 1,70,000 (approx. \$2,300) grant; **East & North-East Regional Winner** – Rs. 60,000 (approx. \$810) grant.
- **Top 100, Marvell M-STEM Scholarship 2026** (national) – USD 2400 (approx. \$3,100) grant.
- **Top 25, Quest Global Ingenium 2026**.

LEADERSHIP

Peer Tutor – Core Engineering Subjects

KIIT

Taught peers and juniors foundational CS/engineering coursework

2023 – 2025

- Ran recurring peer classes over **2 years** covering core engineering subjects for juniors and classmates.

Technical Session Speaker – DL, Quantum & Cloud Computing

KIIT

Sessions for peers, juniors, and seniors

2023 – 2025

- Deep Learning: taught **Explainable AI, Generative AI, Transformers, Encoders, LLMs**, and hybrid models, plus their integration across application domains.
- Quantum Computing: covered foundational quantum mechanics (e.g. **quantum entanglement**) and quantum approaches to **NP-hard problems** such as N-Queens and the Travelling Salesman Problem.
- Cloud Computing: introduced **AWS** fundamentals to peers and juniors.

KIIT AI Society – Technical Session Lead

Aug 2023

Generative AI Symposium Webinar, KIIT University

Bhubaneswar, India

- Delivered a Generative AI session to **200+ participants**; mentored students through AI workshops and hackathons, contributing to **10+ AI projects**.